Marginal Control Quantities in the Single-Affinity Feedback Theory

A short theoretical report

26 August 2026

Basis. This note uses only the finite-time single-affinity reference theory in *Finite-Time Feedback Control with Kinetic Compliance* (22 August 2026). The baseline used for the plots is

$$N = 24, \quad h = 2, \quad \gamma = 0.75, \quad \beta = 4, \quad \theta = 1/2, \quad p_0(n) = \text{Binomial}(24, 0.25),$$

with sensor sizes $q_c \in \{4, 8, 12, 20\}$ and actuation budget $b = 0, \ldots, 24$. All quantities are exact finite-state calculations; no Monte Carlo approximation is used.

1. From susceptibility to marginal control response

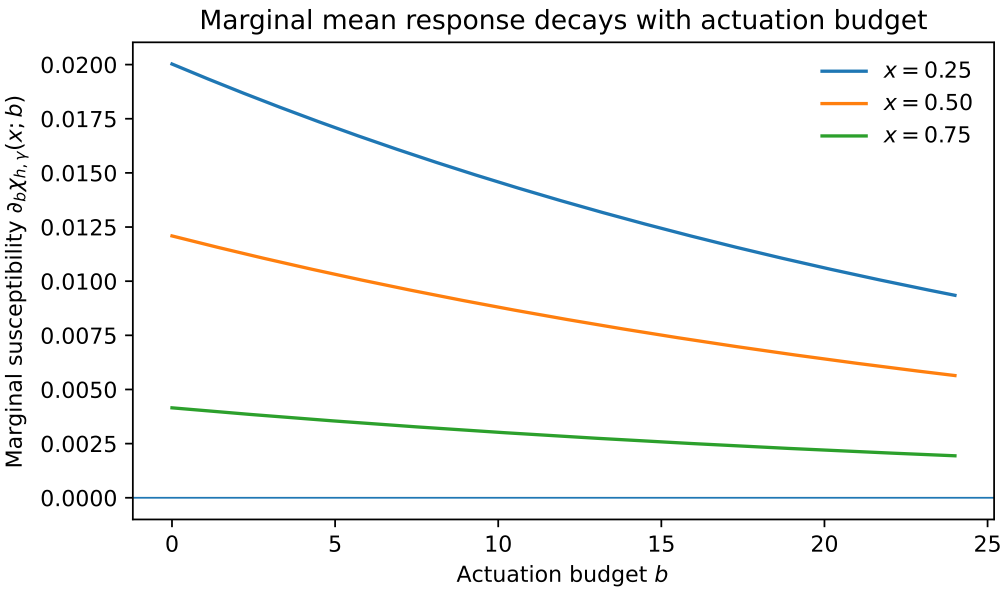
The theory defines the exact ADVOCATE-versus-NoOp susceptibility at target fraction $x = n/N$ as

$$\chi_{h,\gamma}(x; b) = \big[\sigma(h) - x\big]\Lambda_{b,\gamma}, \qquad \Lambda_{b,\gamma} = 1 - \left(1 - \frac{\gamma}{N}\right)^b. \tag{1}$$

The natural local marginal response to increasing the actuation resource is therefore

$$\boxed{M_\chi(x; b) \equiv \frac{\partial \chi_{h,\gamma}(x; b)}{\partial b} = \big[\sigma(h) - x\big]\left[-\ln\left(1 - \frac{\gamma}{N}\right)\right]\left(1 - \frac{\gamma}{N}\right)^b.} \tag{2}$$

For $x < \sigma(h)$ the controller pushes toward the target and $M_\chi > 0$; for $x > \sigma(h)$ it pushes away and the sign reverses. Most importantly, its magnitude decays geometrically with b. Thus the model predicts *diminishing marginal response* analytically: each extra controlled opportunity changes the mean population less than the previous one.



For the ensemble current,

$$J_{c,k} = N\Lambda_{b,\gamma}\sum_n p_k(n) a_n \left[\sigma(h) - \frac{n}{N}\right], \tag{3}$$

the same one-cycle calculation gives $\partial_b J_c \propto \partial_b \Lambda_{b,\gamma}$ when p_k and the policy are held fixed. Hence the current also has diminishing marginal gains with increasing b.

2. Marginal action-to-population information

The theory separately defines the state-local action channel

$$T_\pi(n;b) = I(U_k; n_{k+1} \mid n_k = n), \tag{4}$$

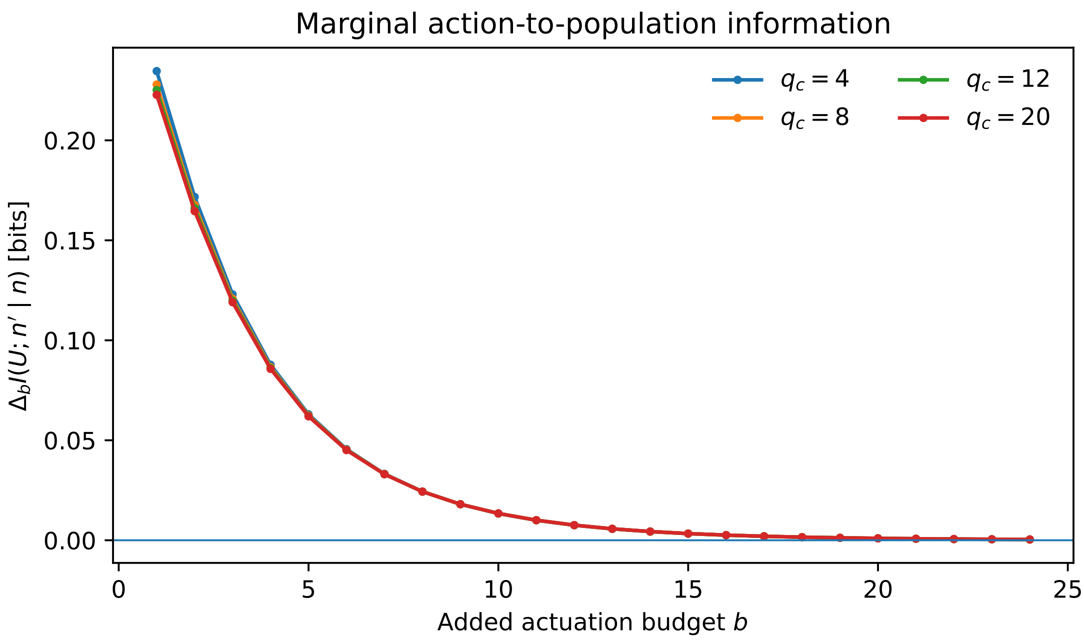
with $Q_0 = I$ and $Q_1 = K^b_{h,\gamma}$. For an occupied ensemble,

$$I_b \equiv I(U_k; n_{k+1} \mid n_k) = \sum_n p_k(n) T_\pi(n;b). \tag{5}$$

Because b is an integer number of controlled opportunities, the operational marginal information quantity is best defined by the exact finite difference

$$\boxed{M_I(b) \equiv \Delta_b I = I_b - I_{b-1}.} \tag{6}$$

It answers a very specific question: *how many additional action-to-population bits become visible when one more controlled opportunity is added?*



The behavior is strongly saturating. For example, at $q_c = 12$ the first added controlled opportunity contributes about 0.225 bits, whereas the 24th contributes only about 3.9×10^{-4} bits. This is sharper saturation than the mean-current response. The reason is structural: once the ADVOCATE and NoOp next-state kernels are already well separated, making the ADVOCATE kernel even stronger adds little new information about which action occurred.

Changing q_c modifies the policy-induced action probabilities a_n and therefore the information landscape, but the qualitative conclusion is robust across the sensor sizes shown: action visibility has a large initial gain and then rapidly diminishing marginal information.

Interpretation. M_χ and M_I are complementary. The first is directional and measures extra mean displacement; the second is unsigned and measures extra distinguishability of controller actions in the next population state. A stronger intervention can still move the population while adding almost no additional action information.

3. Marginal current per additional irreversibility

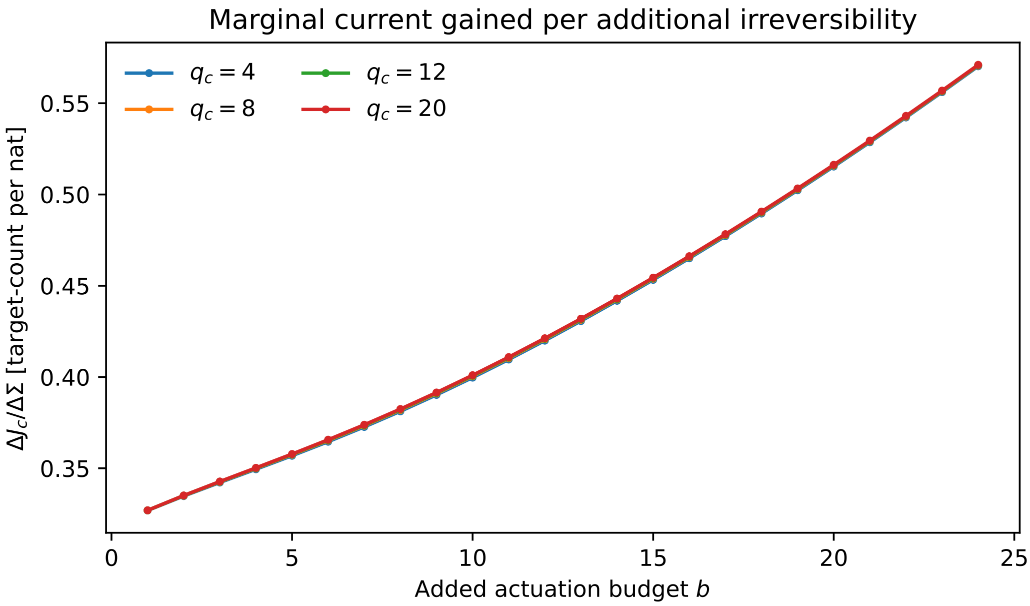
The central finite-time identity of the theory is

$$\Sigma_k = \Delta S_{\mathrm{sys},k} + hJ_{c,k} + I(n_k; Y_k) = D_{\mathrm{KL}}(P_F^{(k)} \| P_R^{(k)}) \geq 0. \tag{7}$$

This makes (J_c, Σ) the natural control-versus-irreversibility pair. If b is the resource being varied, a local slope of this Pareto curve is

$$\boxed{M_{J/\Sigma}(b) \equiv \frac{\Delta_b J_c}{\Delta_b \Sigma} = \frac{J_c(b) - J_c(b-1)}{\Sigma(b) - \Sigma(b-1)}}. \tag{8}$$

It has units of target-count displacement per nat of additional path irreversibility. This is the closest of the three quantities to a thermodynamic marginal efficiency, but it should not be interpreted as a universal bounded efficiency: it is a local trade-off slope for this finite-time reference process.



A nontrivial result appears. $\Delta_b J_c$ decreases with b, but $\Delta_b \Sigma$ decreases even faster. Consequently, $\Delta J_c / \Delta\Sigma$ *increases* over the plotted range. For $q_c = 12$, it rises from about 0.327 target-count/nat for the first actuation unit to about 0.571 target-count/nat for the 24th.

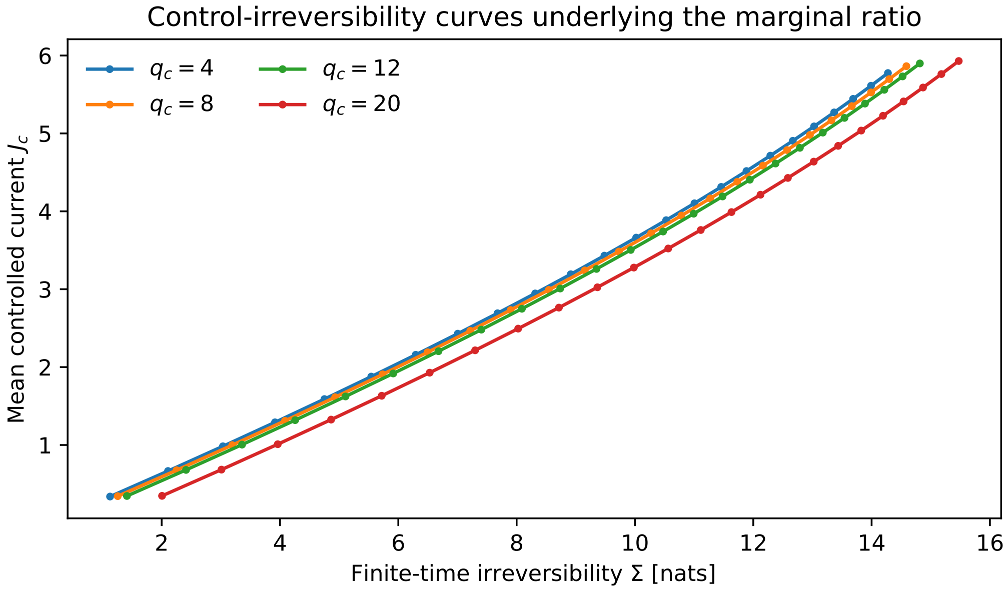
This is not a contradiction with diminishing response. It says two different things:

- extra actuation produces less additional current as b grows;
- but the additional irreversibility required for that extra current falls even more rapidly.

Thus "marginal response per actuation" and "marginal current per irreversibility" need not move in the same direction.

4. Reading the three quantities together

The underlying control-versus-irreversibility curves are shown below. Their slope between successive b values is exactly the finite-difference quantity $M_{J/\Sigma}$ above.



The three marginal quantities separate three questions that were previously mixed together:

$$M_\chi(x;b) = \partial_b \chi \quad : \text{ extra signed mean response per extra actuation,} \tag{9}$$

$$M_I(b) = \Delta_b I(U; n' \mid n) \quad : \text{ extra action information per extra actuation,} \tag{10}$$

$$M_{J/\Sigma}(b) = \frac{\Delta_b J_c}{\Delta_b \Sigma} \quad : \text{ extra current per extra irreversibility.} \tag{11}$$

For the present theory they exhibit different behavior. M_χ decreases smoothly; M_I collapses rapidly toward zero; while $M_{J/\Sigma}$ increases because the irreversibility increment decays faster than the current increment. This is precisely why it is useful not to compress control into a single efficiency number.

The sensor-size curves for $M_{J/\Sigma}$ nearly overlap. At fixed q_c, the sensing mutual information $I(n_k; Y_k)$ is constant with respect to b, so it does not directly contribute to $\Delta_b \Sigma$; q_c influences the slope only indirectly through the state-dependent advocacy probabilities a_n and the resulting final ensemble. This explains why changing sensing resources has a much larger effect on the absolute Σ level than on the local b-slope in this baseline example.

Main theoretical conclusion. The single-affinity model already contains a clean marginal-control story. It predicts diminishing returns of actuation in both mean response and action information, while the thermodynamic trade-off can improve locally because the incremental irreversibility falls faster than the incremental current. Empirically, these three predictions can be tested directly against repeated LLM trajectories without changing the basic observable definitions already used in the project.

Source theory: *Finite-Time Feedback Control with Kinetic Compliance: Control, information, and irreversibility with a single measurable affinity*, 22 August 2026, especially Eqs. (20), (37), (42)–(43), (50), and (79)–(83).